

Convex Functions: Domain

Function:

$$\begin{array}{l} A \subseteq \mathbb{R}^n \\ B \subseteq \mathbb{R} \end{array} \quad f: A \rightarrow B \quad \xrightarrow{\text{Def}} \quad \text{dom } f := A$$

Extended-value function:

$$f: \mathbb{R}^n \rightarrow \mathbb{R} \cup \{-\infty, \infty\} \quad \xrightarrow{\text{Def}} \quad \text{dom } f := \{x \in \mathbb{R}^n : -\infty < f(x) < \infty\}$$

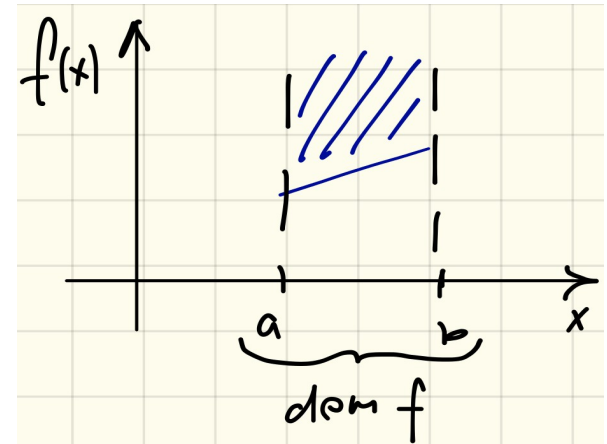
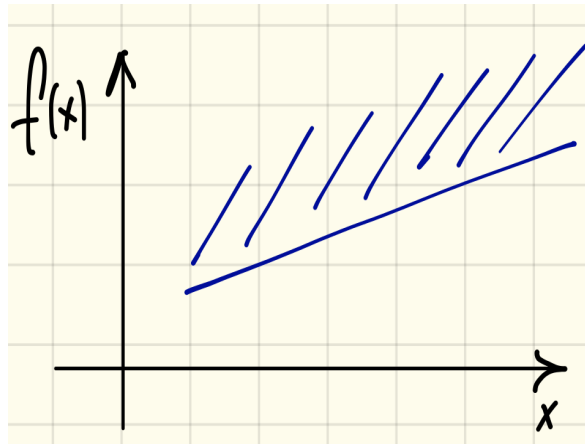
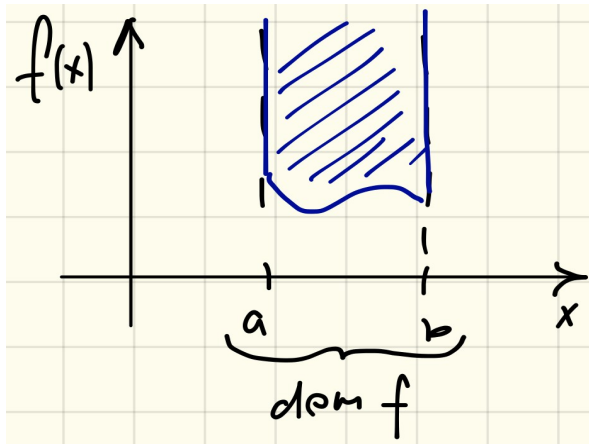


Convex Functions: Epigraph

Epigraph:

$$f: A \rightarrow B$$

$$f: \mathbb{R}^n \rightarrow \mathbb{R} \cup \{-\infty, \infty\} \xrightarrow{\text{Def}} \text{epi } f := \{(x, t) : x \in \text{dom } f, t \geq f(x)\}$$



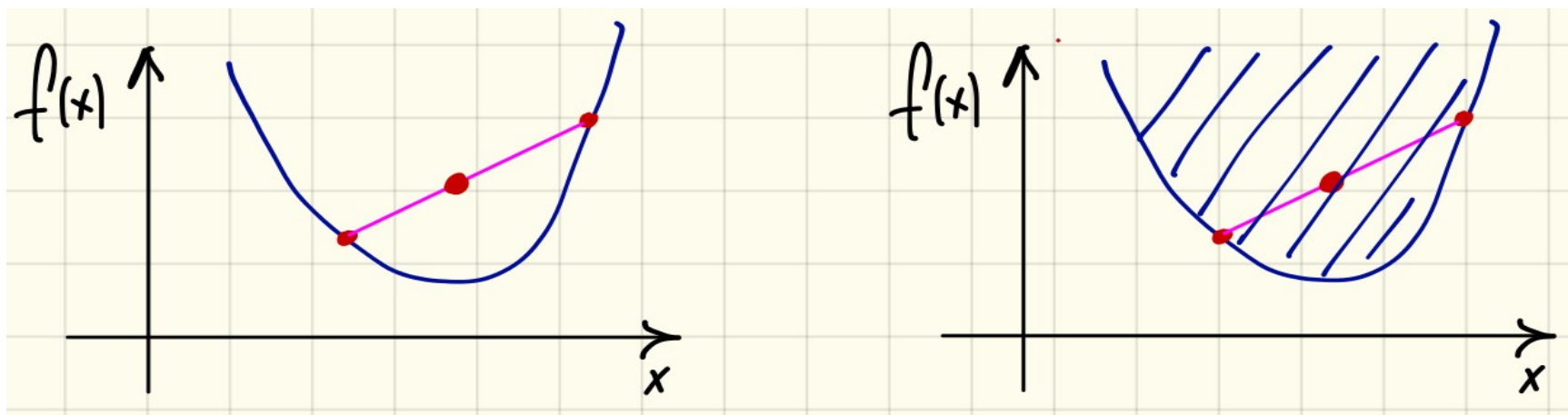
Convex Functions: Definition

Convex function:

$\text{epi } f$ – convex set $\xrightarrow{\text{Def}}$ f – convex function

Convex function:

$$f\left(\sum_{i=1}^n p_i x^i\right) \leq \sum_{i=1}^n p_i f(x^i) \quad \forall n, p \in \Delta^n$$



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Concave function:

$(-f)$ -convex function $\xrightarrow{\text{Def}}$ f -concave function

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Epigraph of a concave function is:

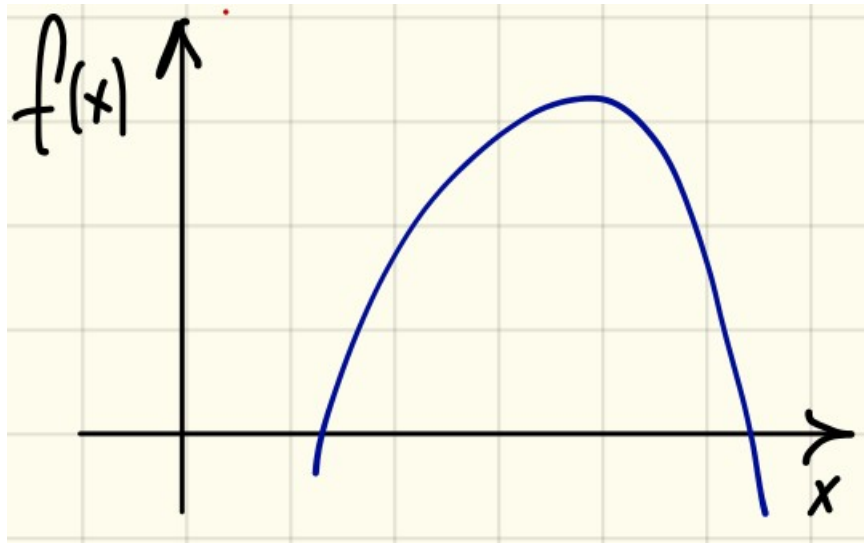
- 1) concave
- 2) convex
- 3) neither concave nor convex in general
- 4) none is correct

Convex Functions: Definition

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$(-f)$ -convex function $\xrightarrow{\text{Def}}$ f -concave function

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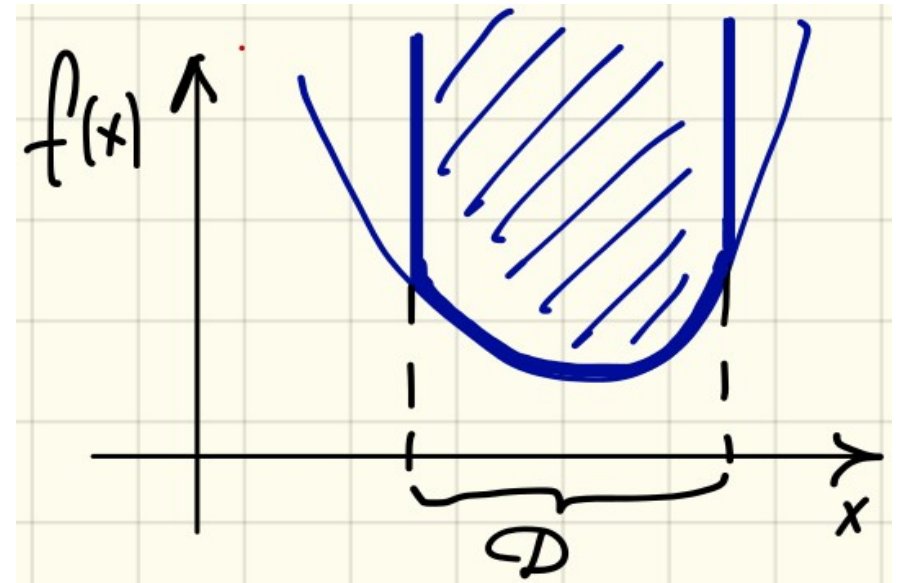
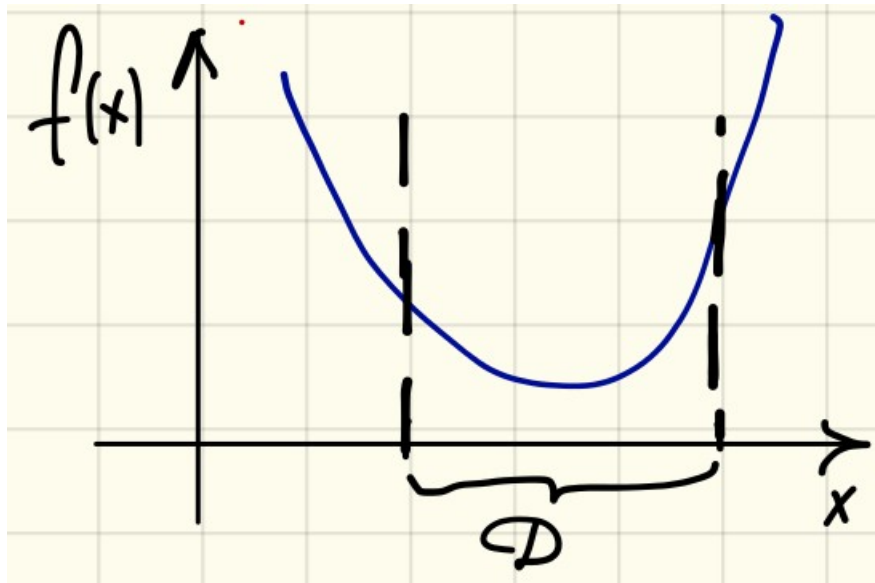
Linear function:

f –convex and concave $\xrightarrow{\text{Def}}$ f –linear

Convex Functions: Examples

Convex function on a convex set of constraints = convex:

$$\begin{array}{l} f: D \rightarrow \mathbb{R} \\ f\text{-convex} \\ D\text{-convex} \end{array} \quad \longrightarrow \quad \tilde{f} = \begin{cases} f(x), & x \in D \\ \infty, & x \notin D \end{cases} \text{ - convex}$$

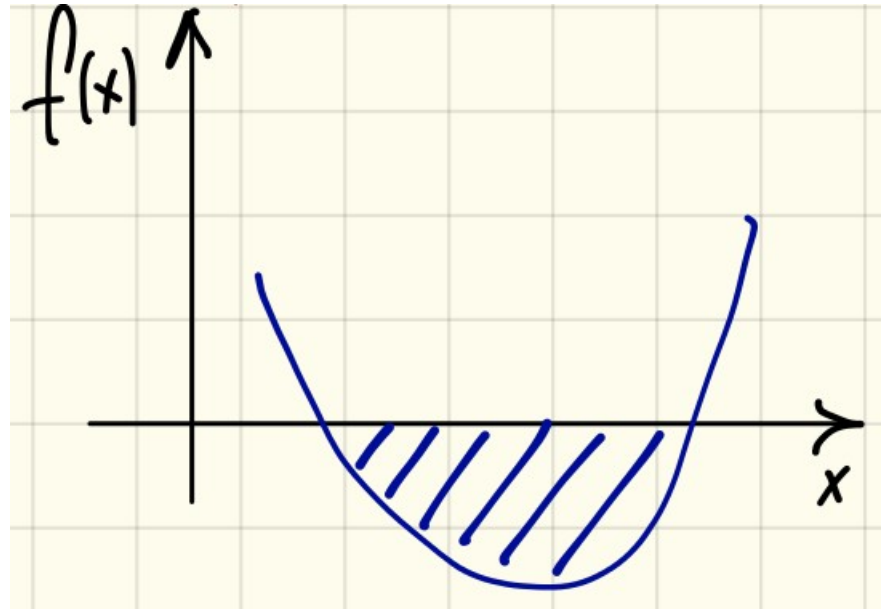


Convex Functions: Examples

Proposition:

$$f: \mathbb{R}^m \rightarrow \mathbb{R} \cup \{\infty\} \text{--convex} \implies \begin{aligned} D &:= \{x: f(x) \leq 0\} \\ D &\text{--convex} \end{aligned}$$

$$D = \mathbb{R}^m \cap \text{epi} f$$



Convex Functions: Examples

Intersection of hyperplanes is convex: $\{x : Ax = b\}$ – convex set

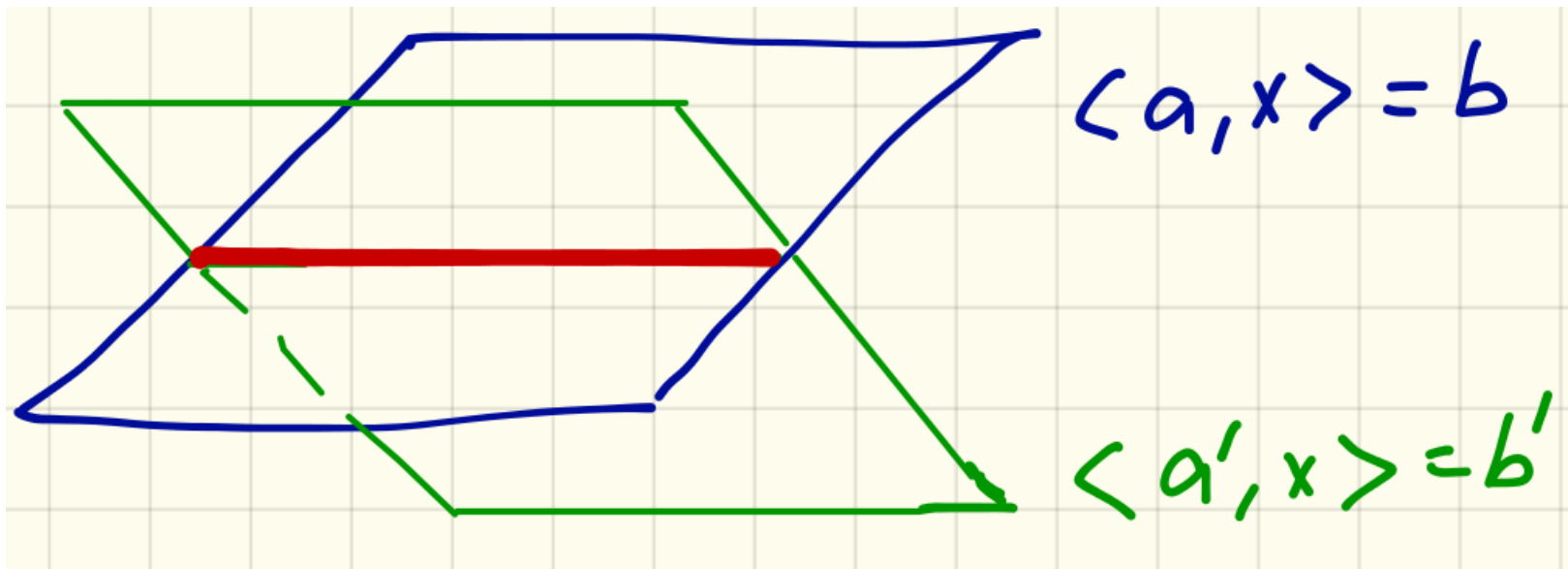
$$p \in \Delta^n \qquad A \sum_{i=1}^n p_i x^i = \sum_{i=1}^n p_i A x^i = \underbrace{\sum_{i=1}^n p_i}_{=1 \text{ by def.}} b = b$$

Here could be your illustration!

Convex Functions: Examples

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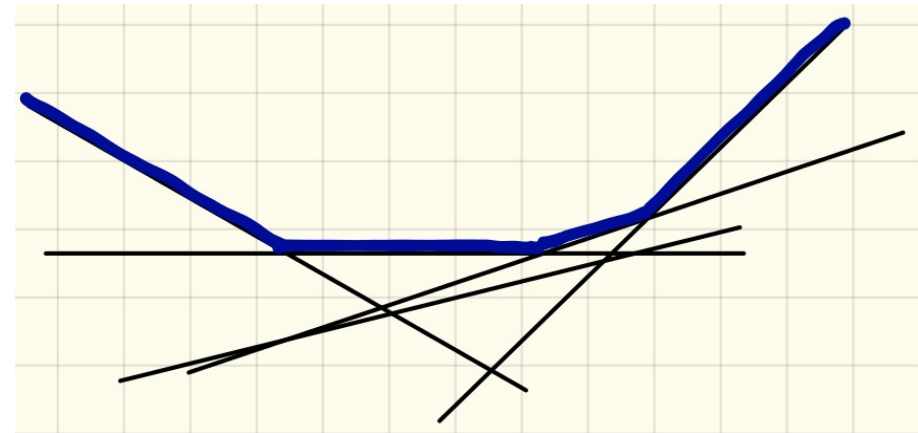
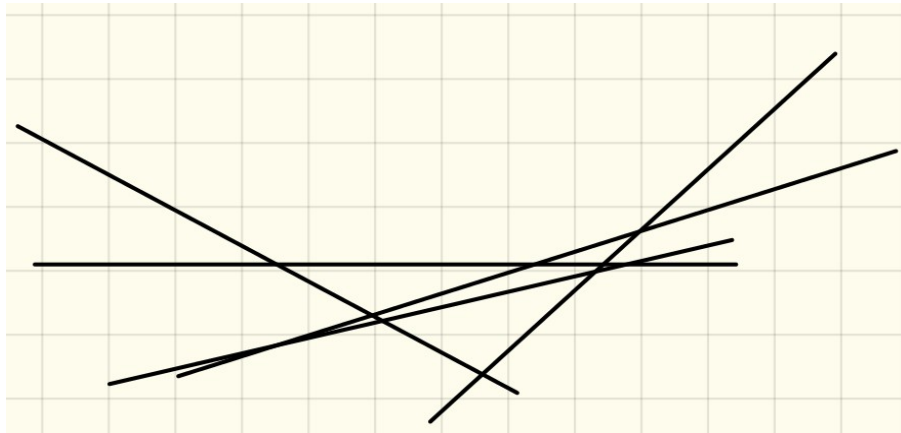
Convex Functions: Examples

Supremum of linear functions is convex:

$$Y \subseteq \mathbb{R}^n$$

$$\begin{array}{ll} f: \mathbb{R}^n \times Y \rightarrow \mathbb{R} & \longrightarrow \quad g(x) = \sup_{y \in Y} f(x, y) - \text{convex} \\ f(\cdot, y) - \text{linear} & \end{array}$$

Proof. Intersection of convex epigraphs is a convex set.



Convex Functions: Examples

Supremum of convex functions:

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$f(\cdot, y)$ – **convex**

- 1) Yes
- 2) No
- 3) I do not know

Draw your picture

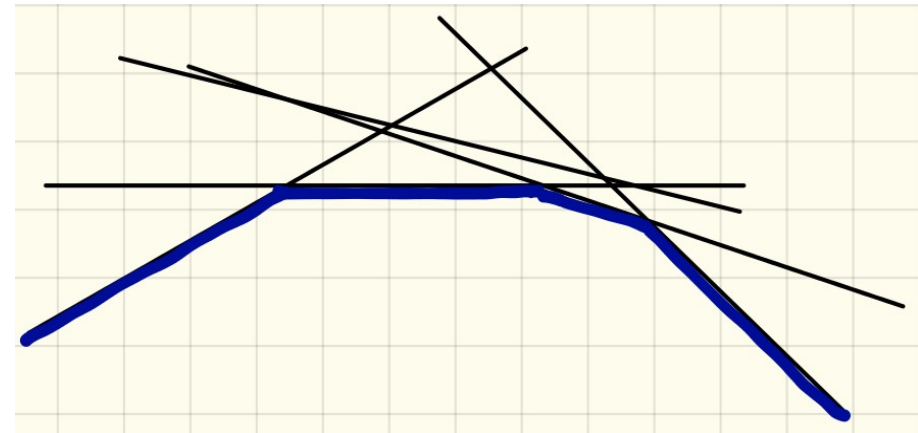
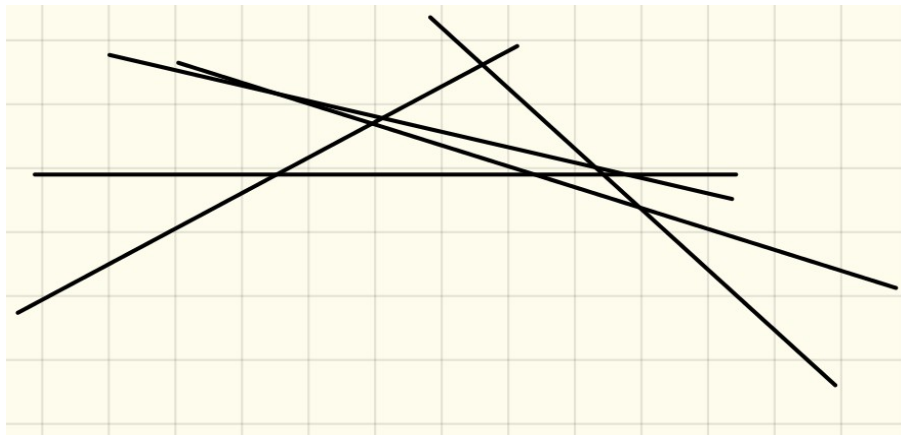
Convex Functions: Examples

Infinum of linear functions is concave:

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$$f: \mathbb{R}^n \times Y \rightarrow \mathbb{R} \quad \longrightarrow \quad g(x) = \inf_{y \in Y} f(x, y) - \text{concave}$$

$f(\cdot, y)$ – linear



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